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Palaeomagnetism of the Stephano–Autunian Lower Tiguentourine formations from stable Saharan craton (Algeria)

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SUMMARY

Palaeomagnetic studies have been carried out in a Stephano–Autunian formation of the Saharan craton. Despite the presence of a strong recent magnetic overprint, some samples give a well-defined characteristic remanent magnetization direction. Moreover, a new approach using the remagnetization circles has been used to confirm the significance of this direction. The obtained Stephano–Autunian pole (35.3°S, 60.3°E) allows a substantial improvement of the apparent polar wander path (APWP) for stable Africa. This APWP clearly shows that the clockwise rotation of Africa, which occured during the Hercynian orogeny, was stopped during the Stephano–Autunian period.

Key words: Africa, apparent polar wander, palaeomagnetism, remanent magnetization.

1 INTRODUCTION

The geodynamic reconstructions for Permo–Triassical times are thwarted by the scarcity of palaeomagnetic data for stable Africa. Indeed, there are few reliable poles for the Lower Permian (Morel *et al.* 1981; Merabet *et al.* 1990). In addition, in the stable parts of this continent, there are often sedimentary breaks during Upper Permian and Lower Triassic times. The previously studied formations of this age, have been unsuitable for palaeomagnetic studies (Ghorabi & Henry 1990).

Nevertheless, geologically well dated and well studied Stephano–Autunian red beds outcrop in the Illizi basin. Such favourable geological conditions led us to investigate these formations in detail. The expected results might reduce the palaeomagnetic data gap and consequently improve the knowledge of Africa's movements for this period.

2 GEOLOGICAL SETTING

The tabular Tiguentourine formation, located in the Saharan platform, corresponds to the uppermost part of the Palaeozoic series of the Illizi basin (Fig. 1). This formation outcrops between the Moscovian of the El-Adeb Larache beds and the cliff of the Mid–Upper Triassic Zarzaïtine sandstones. An important gypsum horizon divides these series into two distinguishable parts. Tiguentourine rocks comprise predominantly shales, muddy siltstones and sandstones, with some interbedded limestones and gypsum fine beds. The shales and the matrix of siltstones and sandstones are mainly a mixture of illite and chlorite, and are almost without kaolinite. The lower boundary of the formation corresponds to a layer with algal concretions. The presence of Estheria and little gastropods (Bertrand Sarfaty & Fabre 1972) at different levels of the lower series, shows that the deposit probably occurred in lacustrine or lagunal conditions; numerous mud cracks correspond to emersion episodes.

The Tiguentourine formation overlaps the well dated El-Adeb Larache formation (Moscovian). The Upper Tiguentourine facies are azoic and could have any age up to Middle Triassic. In contrast, the upper Lower Tiguentourine beds include flat ferruginous nodules, some of them with vertebrates remains: Actinopterygians, Acanthodians and Elasmobranchs. Since Acanthodians disappeared before the end of the Lower Permian, Attar *et al.* (1981) have deduced a Stephano–Autunian age for these Lower Tiguentourine series.

3 SAMPLING AND ANALYSIS PROCEDURES

Because of this well-defined age, the Lower Tiguentourine red beds were chosen for a palaeomagnetic study. 159 samples were collected at 19 sites scattered over an area (27.7°N, 9.0°E; Fig. 1) about 10 km long where the strata are horizontal. Shale hand samples from two sites were too

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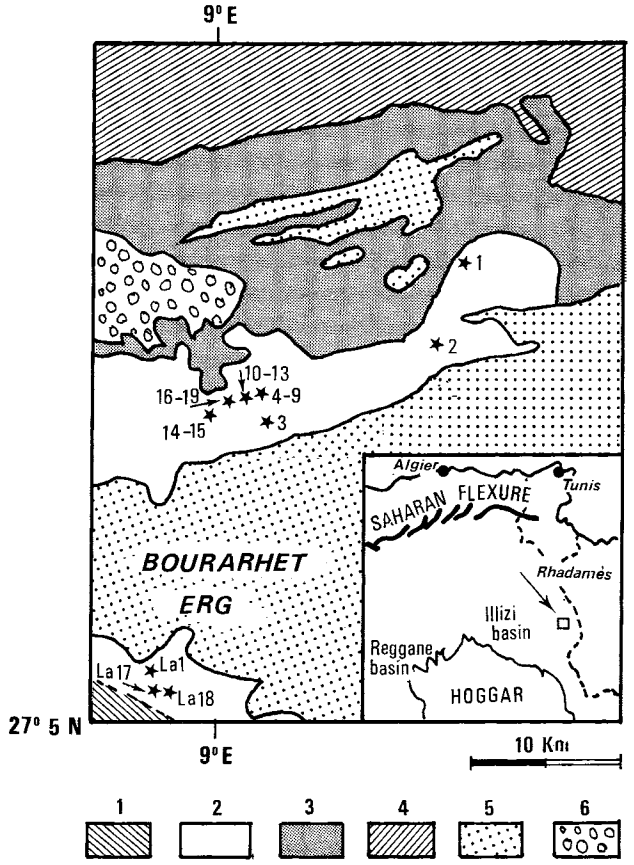


Figure 1. Geological map of the study area and sampling sites (sites La1, La17, La18 from Merabet *et al.* 1990; Henry *et al.* 1992a). (1) El-Adeb Larache (Moscovian). (2) Lower Tiguentourine (Stephano–Autunian). (3) Upper Tiguentourine. (4) Zarzaïtine (Middle–Upper Triassic–Liassic). (5) Erg. (6) Quaternary.

friable to be analysed, but 145 cores were drilled over the remaining 17 sites. One to three specimens were cut from each core. Before any demagnetization analysis, samples were put in a zero field for one month in order to reduce spurious viscous components.

The natural remanent magnetization (NRM) of the specimens so cleaned was analysed using a spinner magnetometer JR4 (GEOFYZIKA, Brno). First, several pilot specimens were subjected to stepwise alternating field and thermal demagnetization in order to get information about their magnetic behaviour. The alternating field treatment did not allow a complete demagnetization, hence, the thermal procedure was applied to all specimens. This last procedure was performed in steps varying from 150 °C at low temperatures to 10 °C at the highest to obtain a more detailed evolution of the remanence. After each heating step, low-field magnetic susceptibility was measured at room temperature on a KLY2 Kappabridge (GEOFYZIKA, Brno) in order to detect any mineralogical phase changes.

Demagnetization data analysis was done using orthogonal diagrams (Zijderveld 1967), stereoplots of the remaining vectors and vectorial differences and the remagnetization circles method (Halls, 1976, 1978). The mean directions of the different components have been computed using Fisher's (1953) and Le Goff, Henry & Daly (1992) statistics.

4 ROCK MAGNETISM STUDY AND PALAEOMAGNETIC MEASUREMENTS

In order to investigate the magnetic mineralogy, we have determined coercivity and unblocking temperature spectra (Lowrie & Heller 1982). Fig. 2(a) shows a spectrum of isothermal remanent magnetization (IRM) acquisition. Few magnetic grains have very low coercivity, and the saturation is not reached at 1.6 T; haematite seems, therefore, to be the dominant magnetization carrier. This is confirmed by thermal demagnetization of this IRM (Fig. 2b) which indicates a Curie temperature of over 650°C. Hysteresis loops show a very dominant effect from paramagnetic and antiferromagnetic minerals. The observed high coervcivity

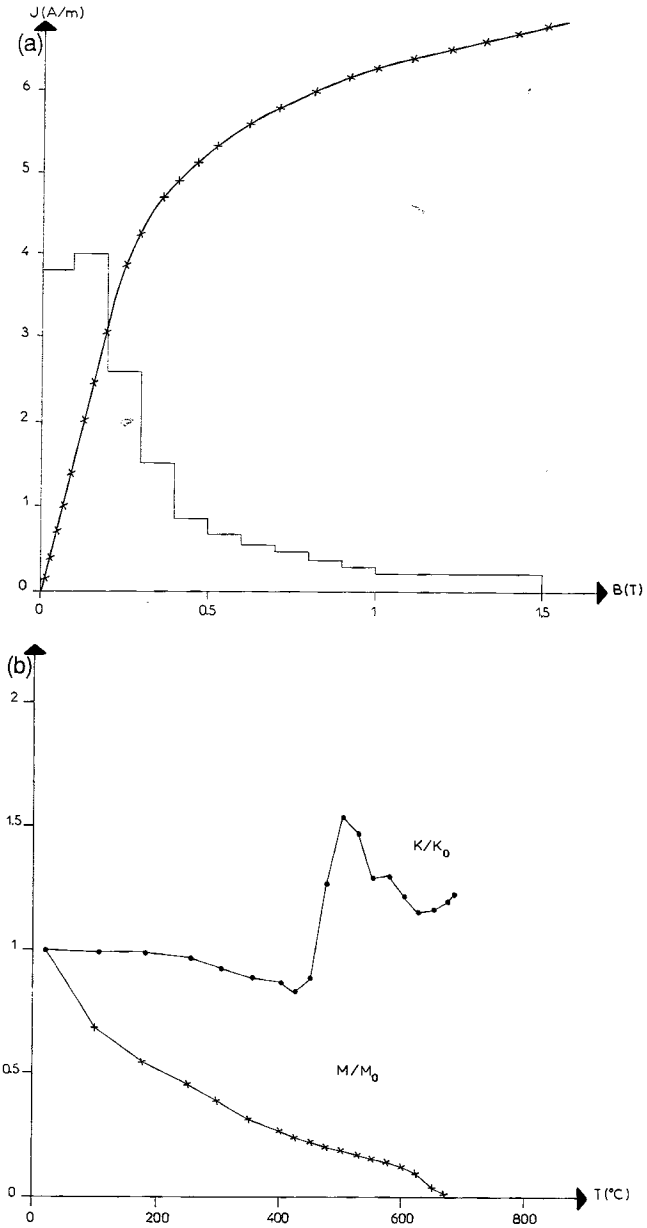


Figure 2. (a) IRM as a function of the applied field and corresponding acquisition spectrum. (b) Normalized magnetic susceptibility (●) and IRM intensity (+) at room temperature as a function of maximum applied temperature.

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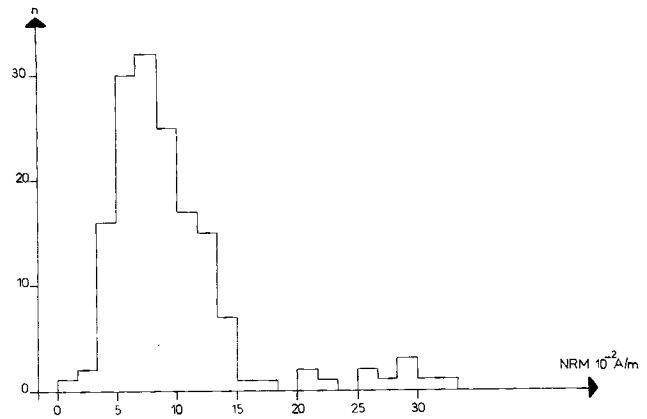


Figure 3. Histogram of NRM intensities.

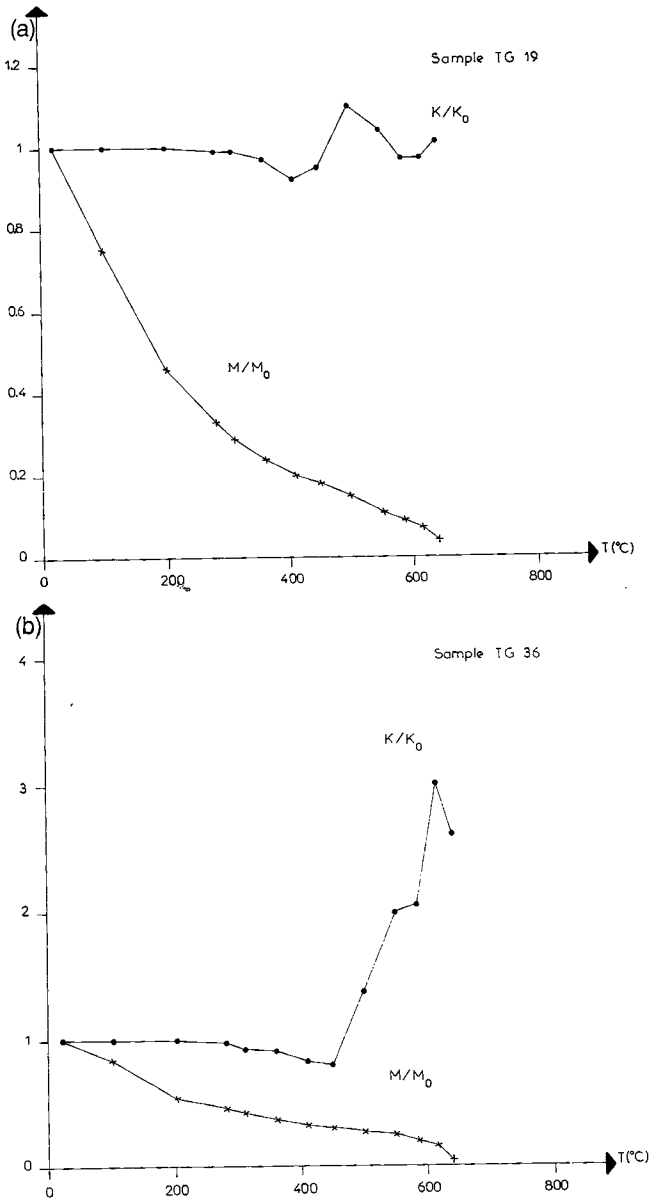


Figure 4. Normalized magnetic susceptibility (●) and remanent magnetization intensity (+) at room temperature as a function of maximum applied temperature. (a) Sample TG19. (b) Sample TG36.

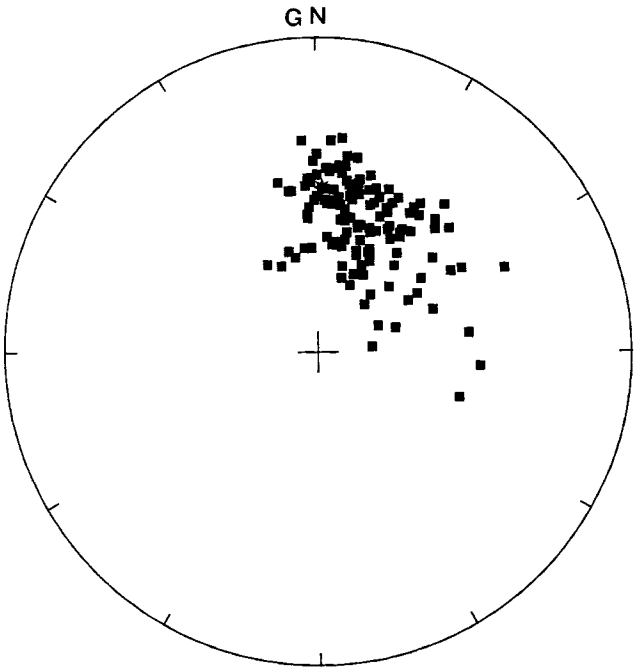


Figure 5. Equal-area plot (filled symbols: lower hemisphere; open symbols: upper hemisphere) of NRM vectors. Star indicates the Earth's present magnetic field direction.

explains the low efficiency of the alternating field demagnetization.

NRM intensities (Fig. 3) generally range from 4 to 310 (in 10^{-3} A m^{-1}), with a mean of 92 10^{-3} A m^{-1}. All specimens show a slight decrease of magnetic susceptibility at room temperature between the 300 °C and 450 °C treatments. At higher temperatures, the magnetic susceptibility generally reaches its initial value or undergoes a moderate increase (Fig. 4a). However, after these thermal steps, some specimens show a much more significant increase of the magnetic susceptibility (Fig. 4b). These variations obviously indicate a mineralogical phase change due to heating. Fig. 4 also presents the NRM thermal demagnetization curves for the same samples. There is no apparent correlation in slope variations between the corresponding curves in Figs 4(a) and 4(b). The same observation can be made for the thermal demagnetization of the IRM (Fig. 2b). Therefore, the mineralogical phase changes do not affect the main carriers of the NRM and IRM.

The majority of the NRM vectors are concentrated around the present magnetic field direction (Fig. 5). The others are spread in a partial girdle between the main cluster and a south-east direction. This arrangement suggests a superimposition of at least two components in the NRM. The magnetization direction for all samples during thermal treatment show a similar evolution towards SE, which also suggests the superimposition of unblocking temperature spectra with at least two components.

5 DISCUSSION

5.1 Recent overprint

The recent overprint has been identified by using only vectorial differences. Indeed, there was no specimen which underwent a complete remagnetization.

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The use of vectorial differences to determine a component requires several conditions. First, in order to avoid a viscous component, the vectorial differences corresponding to temperatures lower than 250 °C were eliminated. Secondly, only the vectorial differences which show a stability in direction for a sufficient number of consecutive thermal treatment steps have been taken into account; the range of temperatures of these vectorial differences varies from specimen to specimen, between 250 to 360 °C and 250 to 615 °C.

Such a selection obviously relies on the fact that there is no overlap of two components at these temperatures, but there is no definite evidence that the component has been well isolated; besides, the significant ellipticity (ratio of k values: 2.6; lengthening of the distribution towards a south-east direction) found in the bivariate statistics (Le Goff *et al.* 1992) is an argument in favour of the existence of such a superimposed component. However, the relative coherence of the results (Fig. 6 and Table 1) from vectorial differences and the coincidence with the mean direction from remagnetization circles indicate that the superimposed component is probably weak. Moreover, the recent overprint directions, all with normal polarity, have a mean direction (D = 16.6°, I = 52.8°) which remains close to that observed in the underlying Moscovian formation (D = 8.3°, I = 46.0°, Henry *et al.* 1992a).

5.2 Stephano–Autunian direction

Stable endpoints were obtained on 28 samples from seven sites (Fig. 6). Since the slight mineralogical changes did not affect the magnetization direction at high temperatures, the 28 mean directions obtained are not the result of parasitic effects due to heating. Table 1 shows the good coherence within each site and between the different sites. In addition, all the remagnetization circles (Figs 7a, c and e) which necessarily contain the NRM components, constitute a girdle including the stable endpoint directions, and moreover intersect in three cases in the same south-east direction (Table 1) as the mean stable endpoints. Finally, the application of bivariate Fisher's statistics (Le Goff *et al.* 1992) with weighting by the number of specimens to all sites shows a moderate ellipticity (k ratio: 1.5). All these arguments confirm that these directions indicate a single component.

The results are summarized in Table 1. The mean direction of the 28 samples, calculated by giving the unit weight to each specimen (D = 138.8°, I = 20.7°, k = 127, α_{95} = 2.3°), is very close to that obtained giving unit weight to each of the seven sites (D = 138.2°, I = 21.2°, k = 170, α_{95} = 4.1°). The determination of this direction takes into account the sites 3, 8 and 19 which contain less than three specimens giving an endpoint. Nevertheless, the results from the remaining four sites are almost the same (Table 1).

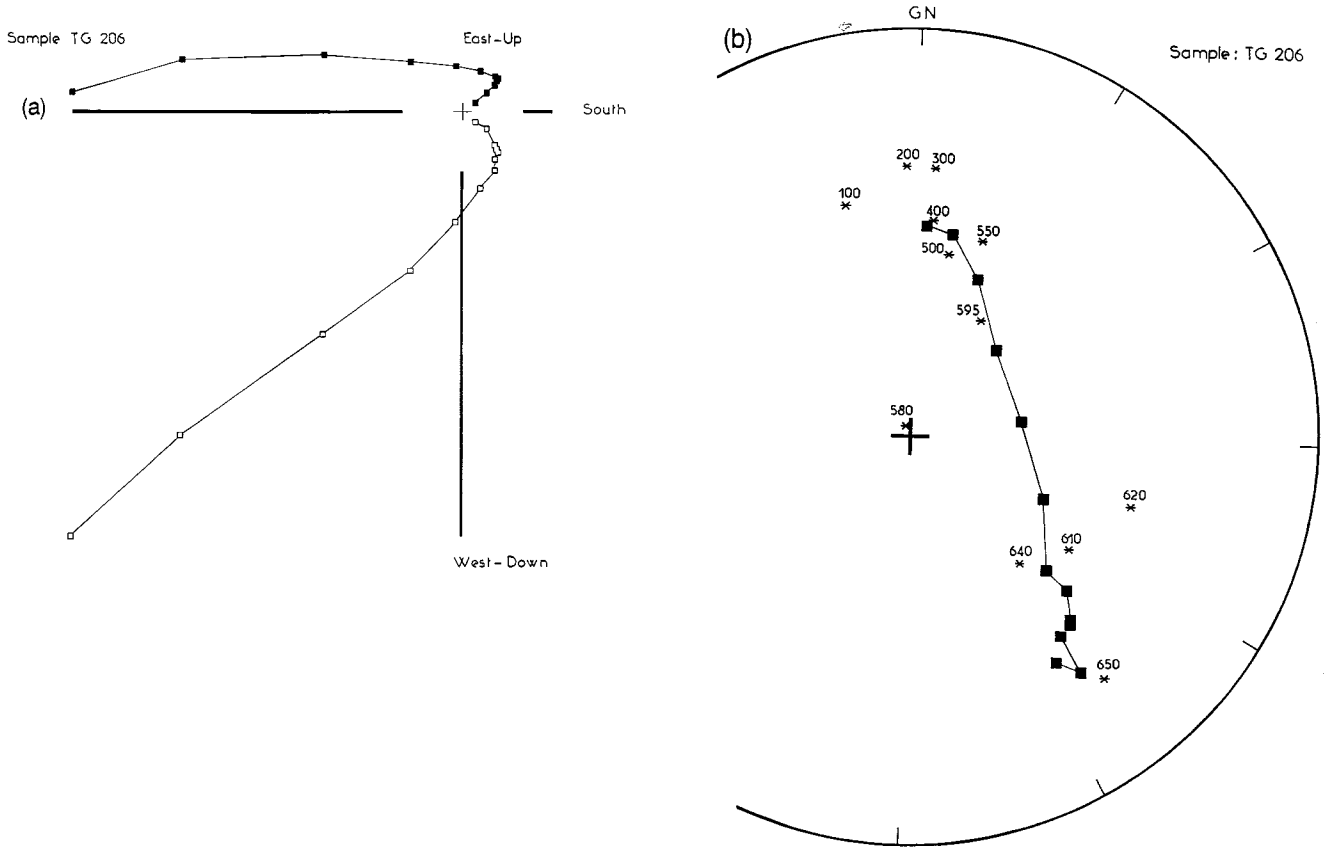
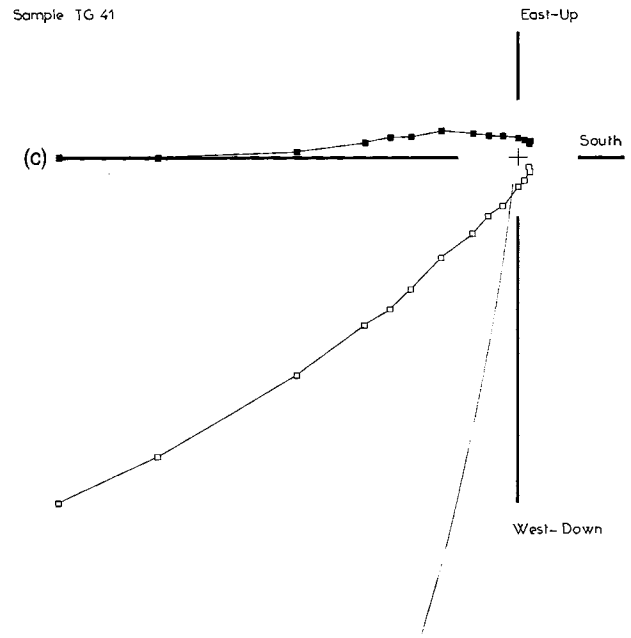
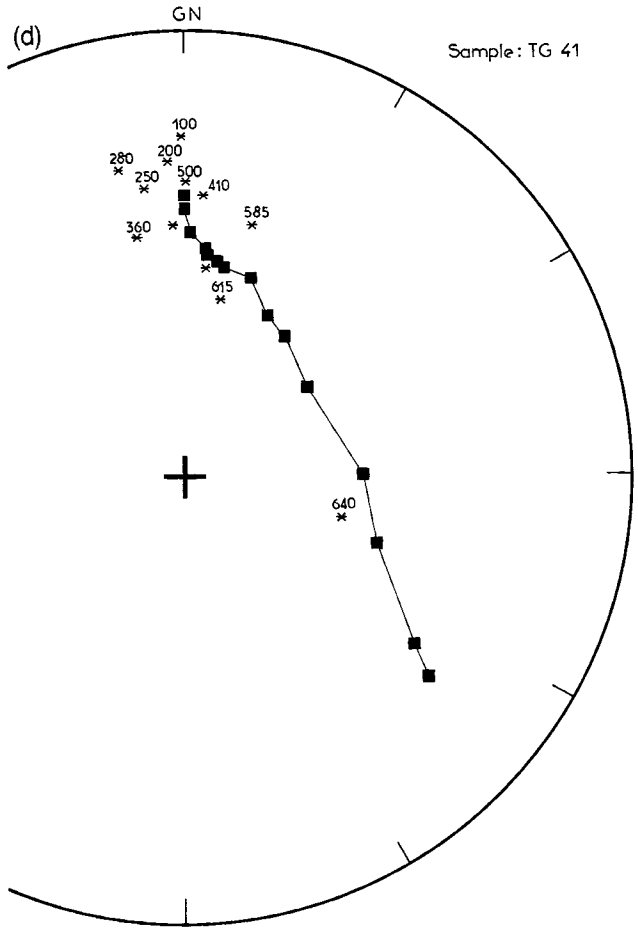
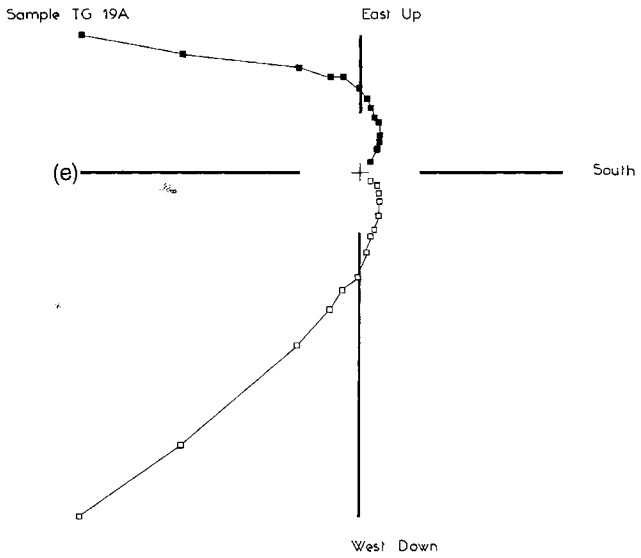


Figure 6. (a, c, e): Orthogonal vector plot (Wilson & Everitt 1963; Zijderveld 1967) (filled symbols: horizontal plane; open symbols: vertical plane). (b, d, f): Equal-area plots (squares: remaining vectors; stars: vectorial differences; see Fig. 5). (a, b) Stable endpoint case (sample TG206). (c, d) High temperature overlapping spectra (sample TG41). (e, f) Complete overlapping (sample TG19A).

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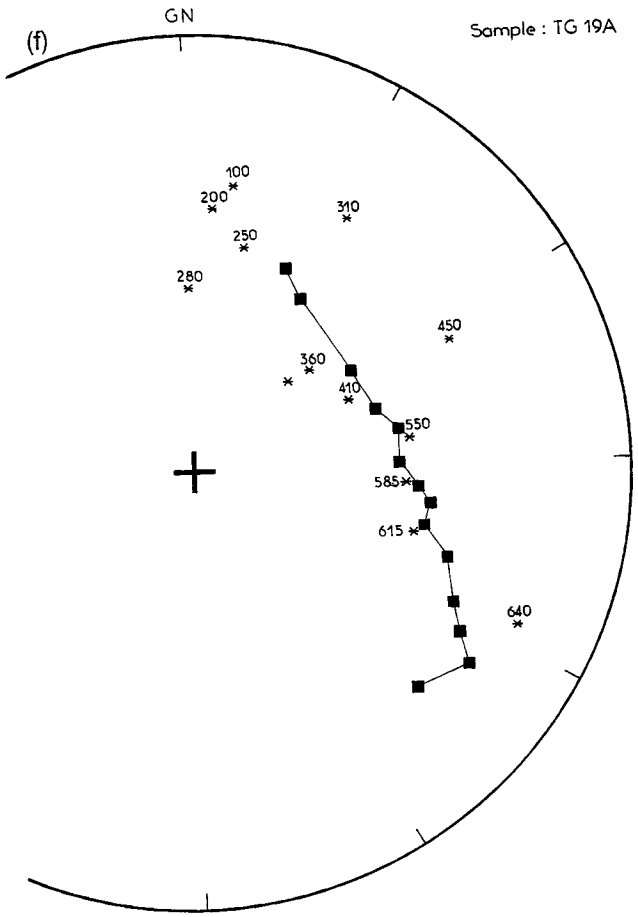


Figure 6. (*Continued.*)

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Table 1. Mean directions (D, I: in degrees) obtained for each site (N: number of samples) and Virtual Geomagnetic Pole VGP: Lat. (°S), Long. (°E). Corresponding Fisher's parameters (k, α_{95}, K, A_{95}: α_{95} and A_{95} in degrees).

Most significant intersection of remagnetization circles

Site	D	I	
4	24	56	}
7	28	60	} Direction close to the recent field
14	02	42	}
8	140	18	}
10	140	22	} S.E. Direction
15	142	20	}

Recent overprint

N	D	I	k	α_{95}	Lat	Long
66	16.6	52.8	67	2.1	74.6	73.3

Stephano-Autunian direction

Site	N	D	I	k	α_{95}	Lat	Long	K	A_{95}
3	2	129.6	25.5						
4	4	137.0	18.1	179	5.2				
8	1	136.2	19.7						
15	3	148.3	17.7	240	5.2				
17	14	139.5	20.4	194	2.7				
18	3	134.0	24.4	91	8.5				
19	1	142.1	21.9						
Mean 28 samples		138.8	20.7	127	2.3	34.6	60.9		
Mean 7 sites		138.2	21.2	170	4.1	34.0	61.1		
Mean 7 sites						33.8	61.4	172	4.0
Mean 4 sites*						35.3	60.2	162	5.5
Mean 10 sites**						35.3	60.3	195	3.2

* VGP derived from the four sites containing more than two specimens giving stable endpoint;
** VGP derived from 10 sites (seven from this study and three from the Stephano–Autunian beds of the El-Adeb Larache formation—Merabet *et al.* 1990, Henry *et al.* 1992a).

The age of the magnetization acquisition remains an important question because of the absence of any element of relative dating such as tilting. During the last two years, systematic palaeomagnetic studies have been performed on sedimentary formations in the Illizi basin. Two of these formations, situated just above (Triassic–Liasic) and below (Moscovian) the formation studied here, yielded good palaeomagnetic directions. The direction from the present formation is intermediate between the Triassic–Liasic (Henry *et al.* 1991) and Moscovian (Merabet *et al.* 1990) ones, but it is closer to the second. As it is different from that of the Permian overprint found in the Moscovian beds (Henry *et al.* 1992a), we can therefore assume that it is primary. Besides, it is quite similar to the Stephano–Autunian direction from beds above the top of the El-Adeb Larache formation (Merabet *et al.* 1990).

5.3 Remagnetization circles

Magnetization directions evolve along great circles during thermal treatments (Creer 1962). Therefore, the most

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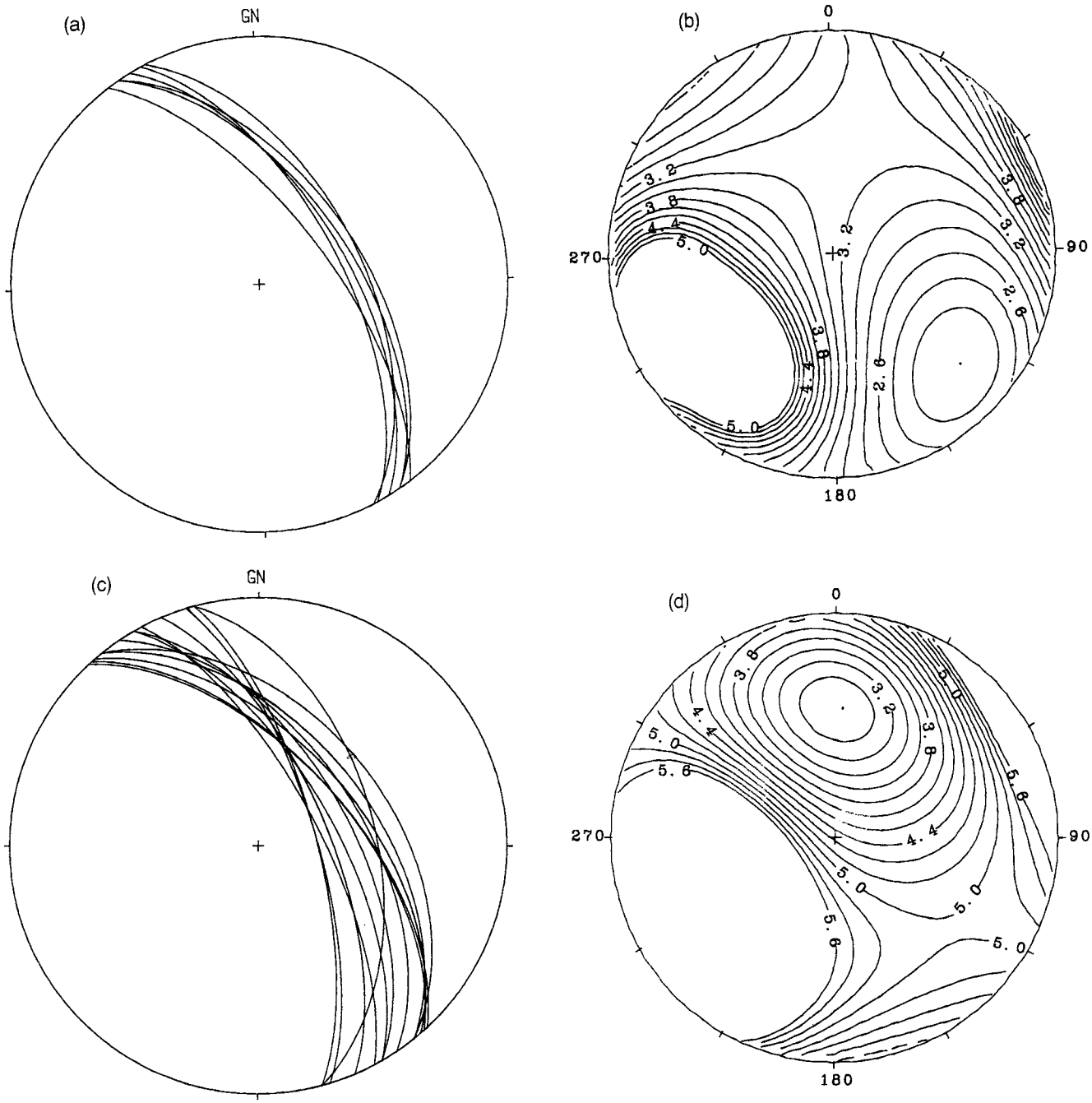


Figure 7. (a, c, e): Remagnetization circles (equal-area plot, lower hemisphere). (b, d, f): Iso-a_{95} curves (equal-area plot, lower hemisphere). (a, b) Example of weak gradient. (c, d) Example with dominant recent overprint. (e, f) Example with dominant Stephano–Autunian direction.

adequate method for analysing the superimposed components could be that of remagnetization circles (Halls 1976, 1978). Three types of unblocking temperature spectra overlap can be observed.

The first type is a partial one characterized by two clusters of vectorial differences, with few intermediate directions between them. The first cluster direction is close to that of the present magnetic field; the second, corresponding to higher temperatures, has a south-east direction and allows the definition of a stable endpoint (Figs 6a and b).

The second type shows a complete overlap at high temperatures which does not allow the isolation of any component. But at the lowest temperatures, the stability of the vectorial differences close to the present field (Figs 6c and d) seems to indicate the absence of overlap.

The third case applies to most of the samples. The vectorial differences are scattered along a great circle and do not reach a stable orientation (Figs 6e and f). Unfortunately, the lack of folding in the studied formation could not yield a clear separation of the different components using

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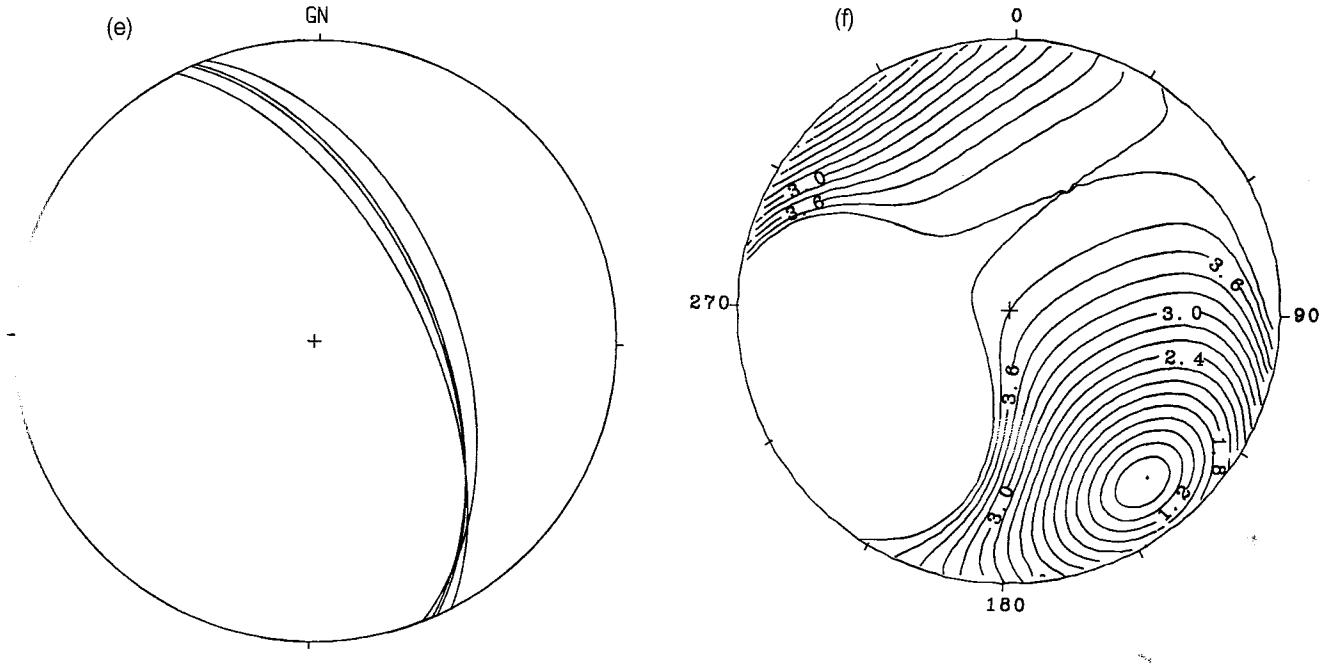


Figure 7. (*Continued.*)

great circles analysis. Indeed, Fig. 7(a) shows an example of the subparallelism between the remagnetization circles of samples belonging to the same site. In such a situation the determination of the best intersection is not always significant (Bailey & Halls 1984).

Since the work of Creer (1962), several methods to determine the best intersection point have been used. McFadden & McElhinny (1988) have proposed one using stable endpoints as constraints to define a sector of great circle within which the estimate of the real direction lies. Using N great circles and taking into account M stable endpoints, an iterative procedure gives the position of N variable-direction unit vectors so that the length of the vector resultant of all $(M+N)$ unit vectors is a maximum. In the present study, the well-grouped stable endpoints are very close to all great circles; results from this method (from 78 great circles and 28 stable endpoints, $D = 139.4°$, $I = 20.7°$, $k = 81$, $\alpha_{95} = 1.5°$) are almost similar to those obtained by direct observations ($D = 138.8°$, $I = 20.7°$; Table 1) which have a very dominant weight. This indicates agreement between endpoints and great circles data, and is therefore a positive information. But the results obtained here from this method clearly depend on stable endpoint orientation (i.e. in practice, it is almost equivalent to assume that one of the components of magnetization in the great circle cases corresponds to stable endpoints, and this could therefore be considered as a circular argument); they do not really yield further information (such as results from sites where there are no stable endpoints or an undeniable argument showing that the stable endpoint is one of the components of magnetization in the great circle case).

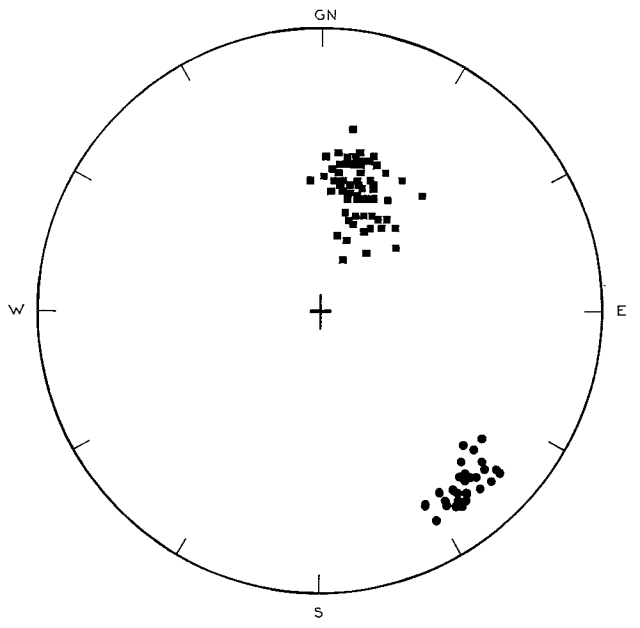


Figure 8. Characteristic remanent direction (square: recent overprint; circles: Stephano–Autunian magnetization) see Fig. 5.

In order to obtain results independent of the stable endpoint orientation (Surmont 1989a and b), McFadden & McElhinny's method has been applied, restricted to great circles data ($M = 0$). An α_{95} value can be determined for the position of the N vectors corresponding to the maximum resultant vector (minimum α_{95}): here, from 78 great circles, $D = 354.7°$, $I = 37.6°$, $k = 106$, $\alpha_{95} = 1.6°$. Nevertheless, such an α_{95} value is obviously underestimated and cannot be compared with an α_{95} value obtained from stable endpoints. Moreover, an area of the projection sphere around the

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maximum resultant vector can include directions that correspond to α_{95} values very close to that of the minimum α_{95} (here, from the 78 great circles, almost similar k and α_{95} values can be obtained from all the directions close to the great circles, and the direction for minimum α_{95} value is not significant, although the α_{95} value is low). This underlines the lack of statistical significance of this minimum α_{95} angle (indeed any α_{95} value obtained by this method is a kind of precision parameter, and to avoid confusion, this value will be called a_{95} in the following discussion).

This is the reason why it may be interesting to investigate, in each particular case, the evolution of the a_{95} values according to the distance from the best estimated direction. Indeed, a weak variation of these a_{95} values implies a low significance of the maximum resultant direction. This is of particular importance in the present study because of the proximity of remagnetization circles. The aim of such an investigation is to obtain a simple and practical method (presently only qualitative) to visualize the importance of the a_{95} gradient around the resultant direction. So, the a_{95} value has been determined for numerous directions of the half-space (in practice using a grid with 2° spacing), and the curves of iso-a_{95} values have been drawn in the stereographic projection.

In order to have the most coherent results for an eventual 'primary' component, this graphical method has been applied only to samples from the same sites (in 12 sites; four remagnetization circles were at least obtained from each site). So, two main types of figures appear. For six sites, differences in a_{95} value along the great circles are very low, and thus no significant results can be obtained from the remagnetization circles method (Figs 7a and b). In the other cases, low gradients of a_{95} values are mainly present in a restricted area around the mean direction (an area therefore corresponding to a kind of 'confidence zone'), and then the gradient increases significantly around this zone. It can be noticed that these 'confidence zones' are restricted to two different directions (Table 1) according to the sites. The first one (Fig. 7c and d) is close to the recent field (three sites, mean direction: D = 15.1°, I = 53.3°) and the other (Figs 7e and f) lies along a south-east direction with a moderate dip (three sites, mean direction: D = 139.7°, I = 20.0°).

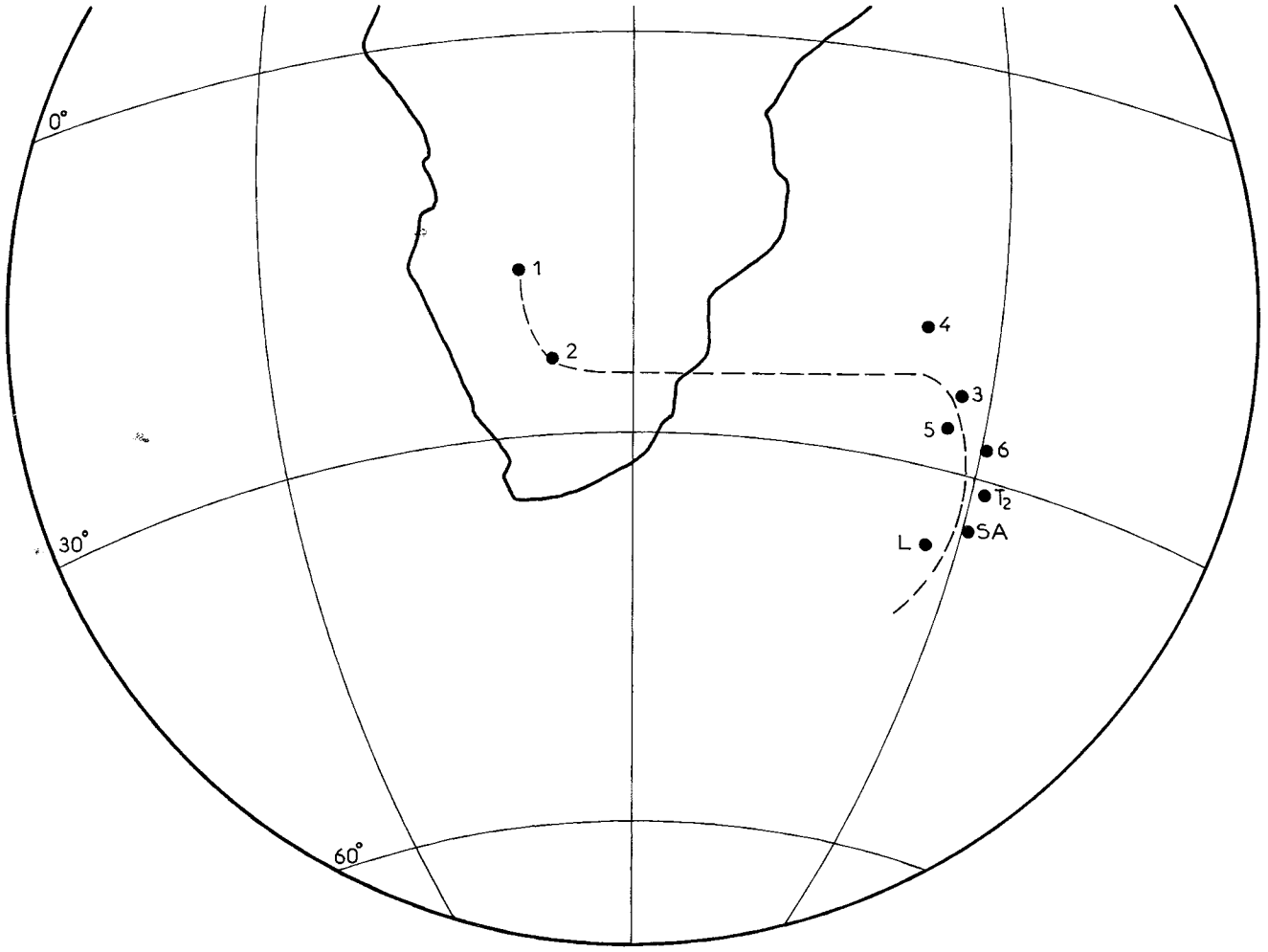


Figure 9. Upper Palaeozoic VGPs and corresponding APWP for stable Africa: (1) Famenian of Ben Zireg (Aïfa *et al.* 1990); (2) Tournaisian remagnetization at Ben Zireg (Aïfa *et al.* 1989); (3) Upper Namurian–Lower Moscovian of Hassi Bachir (Daly & Irving 1983); (4) Lower Moscovian at Aïn Ech Chebbi (Daly & Irving 1983); (5) Moscovian of El-Adeb Larache (Merabet *et al.* 1990; Henry *et al.* 1992a); (6) Stephano–Autunian of Abadla (Morel *et al.* 1981). (L) Stephano–Autunian pole of El-Adeb Larache (Merabet *et al.* 1990; Henry *et al.* 1992a); (T2) present study; (SA) combined pole (L + T2).

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In conclusion, two main points can be underlined. The first relates to the method of great circles analysis (McFadden & McElhinny 1988). This last method appears useful even when the great circles are subparallel, and drawing of iso-a_{95} value indicates the significance of the results. The second point concerns the practical results obtained here by application of this graphical method. The plot of iso-a_{95} curves permits the definition of two clear preferential directions; such an arrangement cannot be due to a mere chance, but reflects the existence of two components. However, a surprising fact is the dominance of one or the other component according to the site. The dominance of the direction close to the recent field appears in sites where the scattering in declination of the south-east direction is apparently much more important than in the other sites (in fact, it often corresponds to a significant uncertainty in the position of the great circle, except for the part of the circle close to the recent field; moreover, some great circles converge along this south-east direction—see Figs 7c and d).

5.4 Palaeopoles

The obtained directions yield two new palaeopoles. The first pole, *T*1, at Lat. = 74.6°S, Long. = 73.3°E, is that of the recent overprint. The second pole, *T*2, at Lat. = 33.8°S, Long. = 61.4°E, K = 172 and $A_{95} = 4.0°$ (Table 1), represents a new palaeomagnetic datum which will improve the definition of the Stephano–Autunian pole of the African curve (Fig. 9). The *T*2 pole is very close to that of the Stephano–Autunian beds (*L*) outcropping at the top of the El-Adeb Larache formation (Lat. = 38.5°S, Long. = 57.5°E, $K = 1459$ and $A_{95} = 2.5°$; Merabet *et al.* 1990). These poles (*T*2 and *L*) were determined from different neighbouring areas covering a large period of the Stephano–Autunian. When combined, they yield a datum (SA) more representative of the Stephano–Autunian age (from 10 sites, Lat. = 35.3°S, Long. = 60.3°E, $K = 195$ and $A_{95} = 3.2°$). This SA pole is near to the single other Stephano–Autunian pole for the Saharan craton at Abadla (Lat. = 29°S and Long. = 60°E; Morel *et al.* 1981).

These Stephano–Autunian results provide important constraints on the (APWP) of stable Africa, since there is no data of upper Permian and Lower Triassic age (Fig. 9).

GEODYNAMIC IMPLICATIONS

The previous palaeomagnetic studies from the stable Saharan craton for the Late Devonian–Early Carboniferous (Ben Zireg–Aïfa, Feinberg & Pozzi 1990) and the Upper Carboniferous (Hassi Bachir, Ain Ech Chebbi—Daly & Irving 1983; El-Adeb Larache–Merabet *et al* 1990; Henry *et al* 1992a) showed that between Tournaisian and Moscovian periods the general trend of the latitudinal movement of Africa was a clockwise rotation of about 25°. But the end of this rotation was not well dated because of the low reliability of the Stephano–Autunian data. This SA pole and the Abadla one allow the deduction that the rotation of Africa was stopped during the Stephano–Autunian period, probably because of continental collision. Evidence of such Late Hercynian orogeny exists in Mauritania (Lécorché, Dallmeyer & Villeneuve 1989) and the southern Appalachian mountains (Hatcher & Odom 1980), but not in southern Europe (where there was, rather, extension at this period); the Pange *A*2 (Van der Voo 1981) hypothesis therefore fits well with geological and geophysical data (Henry *et al.* 1992a and b), in contrast to the Pangea *B* (Irving 1977) model. Therefore, the last phases of the formation of Pangea *A*2 probably occurred as early as this Stephano–Autunian period.

CONCLUSION

The lower Tiguentourine formation, which has a well established Stephano–Autunian age, is located in the stable Saharan craton; in such conditions, palaeomagnetic investigations had some interest, to better the knowledge of Africa's movements.

Despite the presence of a strong recent magnetic overprint, the primary magnetization has been isolated in seven sites. The corresponding *T*2 pole is coherent with the few previous data. This new contribution enables us to improve the African APWP for this key period of the evolution of Pangea. Indeed, this result permits the dating of the end of the Carboniferous clockwise rotation of the Gondwana, and consequently to show that the last phases of the formation of Pangea *A*2 probably occurred during Stephano–Autunian times.

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